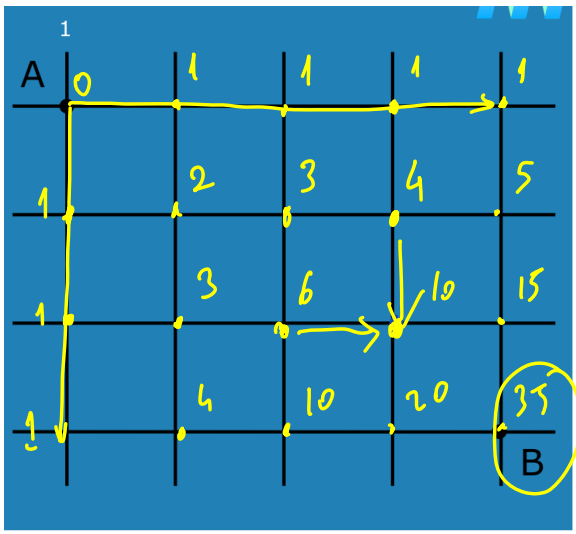


$F(i,j)$ = # shortest paths from $(1,1)$ to (i,j).

$$F(i,j) = F(i-1,j) + F(i,j-1)$$

$F(1,1) = 0$

$F(1,j) = F(i,1) = 1$ $\quad i>1$, $j>1$

$F(4,5) = 35$ is the answer.

Time complexity $= \Theta(nm)$ $\quad n$ is the # rows, m // // # colums.

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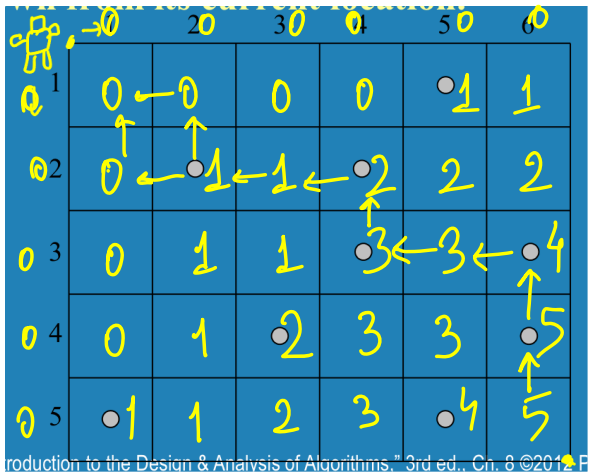
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$$\binom{7}{3} = \binom{7}{4}$$

$$= \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} = 35$$



$$F[i,j] = \max \begin{cases} F[i-1,j] + c_{i,j} \\ F[i,j-1] + c_{i,j} \end{cases}$$

$$= \max \{ F[i-1,j], F[i,j-1] \} + c_{i,j}$$

$$F[0,j] = F[i,0] = \emptyset$$

Time complexity = $\Theta(m.n)$

Change-making problem:

We have n amount of money.

We have coin denominations: $d_1, d_2, \ldots d_m$

$1, \ldots$

Goal is to change n amount of money using min number of coins.

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$d_1 = 1$
$d_2 = 2$
$d_3 = 5$
$d_4 = 10$
$d_5 = 20$
$d_6 = 50$

$5\ d_6$
$1\ d_5$
$1\ d_4$
$1\ d_2$
$+\ 1\ d_1$

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9 coins.

$n = 6 \quad d_1 = 1 \quad d_2 = 3 \quad d_3 = 4 \quad 2 \times d_2$ answer = 2//

$F(n)$ = min # coins to change n.

$$F(n) = \begin{cases} F(n-d_1)+1 & \text{if we we } d_1 \text{'s} \\ = F(n-d_2)+1 & \text{// // } d_2 \\ = F(n-d_3)+1 & \text{// // } d_3 \\ = F(n-d_m)+1 & \text{// // } d_m \end{cases} \Bigg\} \text{choose min}$$

$$F(n) = \min_{d_j \le n} \{ F(n - d_j) \} + 1$$

$F(1) = 1 \qquad F(0) = 0$

$d_1 = 1 \quad d_2 = 3 \quad d_3 = 4 \qquad n = 6$

$F(1) = 1$

$F(2) = \min \{ F(2-1) \} + 1 = 2$

$F(3) = \min \{ \underbrace{F(3-1)}_{2}, \underbrace{F(3-3)}_{0} \} + 1 = 1$ (d_2)

$F(4) = \min \{ \underbrace{F(4-1)}_{1}, \underbrace{F(4-3)}_{1}, \underbrace{F(4-4)}_{0} \} + 1 = 1$

$F(5) = \min \{ \underbrace{F(5-1)}_{1}, \underbrace{F(5-3)}_{2}, \underbrace{F(5-4)}_{1} \} + 1 = 2$

$F(6) = \min \{ \underbrace{F(6-1)}_{2}, \underbrace{F(6-3)}_{1}, \underbrace{F(6-4)}_{2} \} + 1 = 2$ (d_2)

the coins that are selected are : d_2, d_2

Time complexity = $O(n.m)$

$V[i,j]$ ⇒ max value that we can obtain with first i items and capacity of j.

Goal ⇒ $V[n, W]$

$V[i,j] = V[i-1, j]$ if we don't pick the ith item.

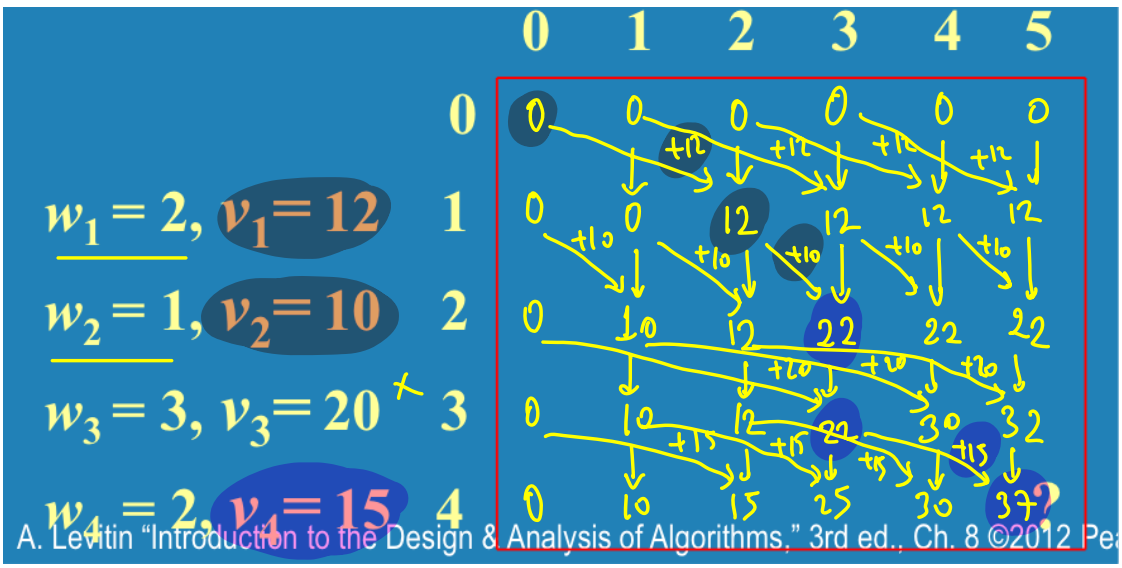
$v_i + V[i-1, j-w_i]$ if we pick the ith item.

so $V[i,j] = \max\{ V[i-1, j-w_i] + v_i, V[i-1, j] \}$ if $j \geq w_i$

$V[i-1, j]$ if $j < w_i$.

$V[0,j] = 0$ $\quad$ $V[i,0] = 0$

j



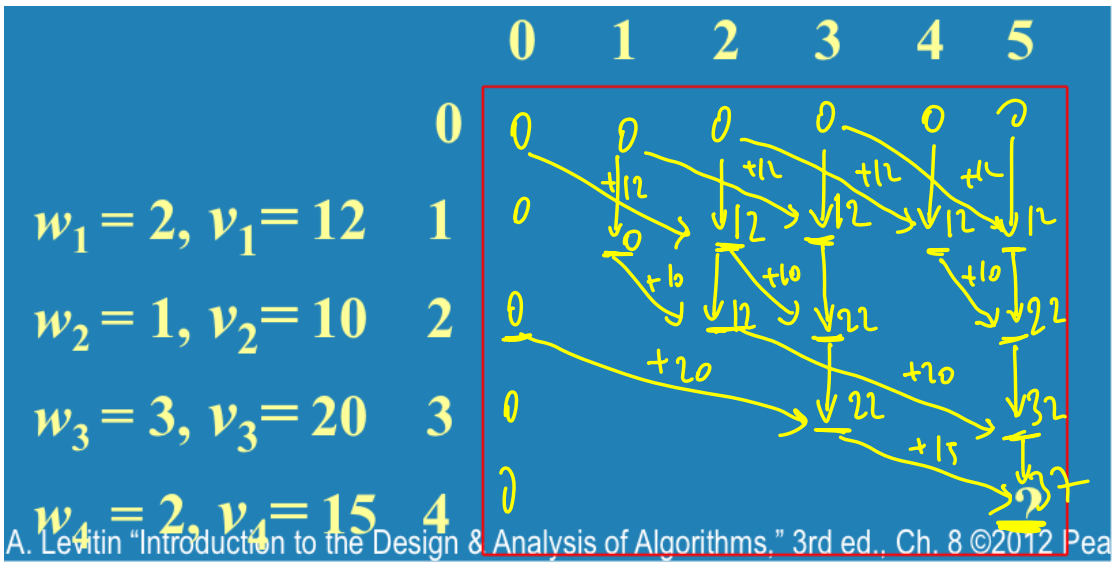
We can find the used items by going backwards.

items 4, 2, 1 are used.

The complexity of the algorithm:

Time complexity = $\Theta(nW)$ $\quad W = 2^b$ b is the length of the binary representation of W.

$\Theta(n \cdot 2^b)$



Memory Function (recursive calls)

Knapsack Problem with infinite copies of items

W: capacity n items: v_i, w_i.

Example: Knapsack of capacity $W = 5$		
item	weight	value
1	2	\$12
2	1	\$10
3	3	\$20
4	2	\$15

capacit

→ For this instance, the result would be:
5 × item 2 ⇒ 50

1	2	\$25
2	1	\$10
3	3	\$45
4	2	\$15

W = 5

1 item 3, 1 item 1 ⇒ 70

Let's set up a recurrence relation for this version of Knapsack Problem:

$F(i)$ = max value with capacity of i

$$F(i) = \max \begin{cases} F(i - w_1) + v_1 \\ F(i - w_2) + v_2 \\ F(i - w_3) + v_3 \\ F(i - w_4) + v_4 \end{cases}$$

goal is to find $F(W)$.

$$F(i) = \max_{w_j \le n} \{ F(i - w_j) + v_j \}$$

$$F(0) = 0$$

Example: Knapsack of capacity $W = 5$			
item	weight	value	
1	2	\$12	
2	1	\$10	
3	3	\$20	
4	2	\$15	capacit

item

$F(1) = \max \{ F(1-1) + 10 \}$

$= 10$

item

$F(2) = \max \{ F(2-1) + 10,$
$F(2-2) + 12,$
$F(2-4) + 15 \}$

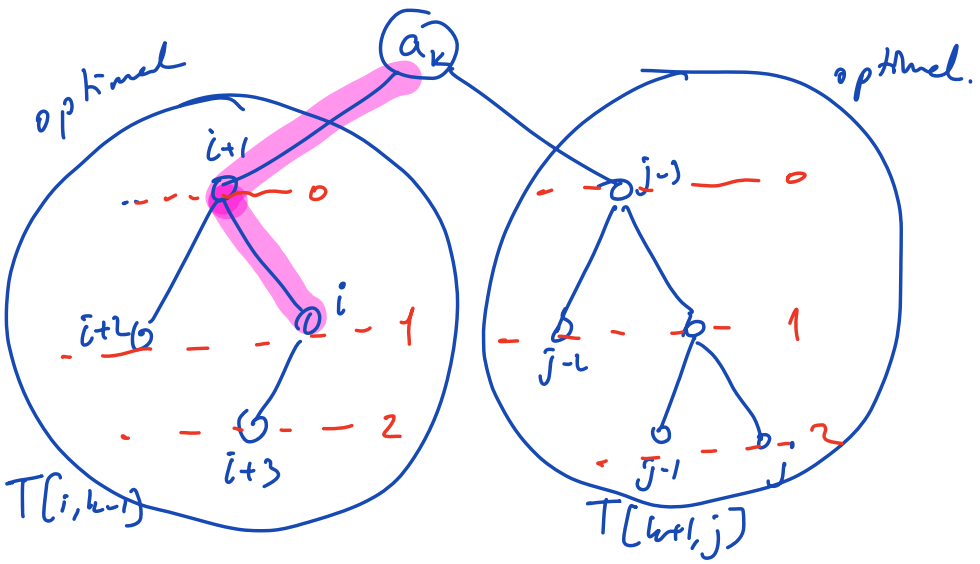
$= 20$

$F(3) = \max\{ F(3-2)+12,\ F(3-1)+10,\ F(3-1)+20,\ F(3-4)+15 \}$
item 2 $= 30$

$F(4) = \max\{ F(4-2)+12,\ F(4-1)+10,\ F(4-1)+20,\ F(4-4)+15 \}$
item 2 $= 40$

$F(5) = \max\{ F(5-2)+12,\ F(5-1)+10,\ F(5-1)+20,\ F(5-1)+15 \}$
item 2 $= 50$

Time complexity: $O(W \cdot n)$



node i: Level of i in $T[i, k-1] = 1$

$p_i \cdot 1$

In the whole tree, Level of i (in $T[i,j]$) $= 1+1 = 2$

$p_i \cdot 2$

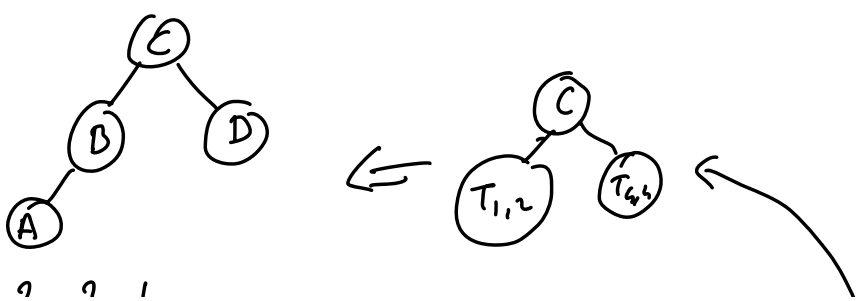
$\sum_{s=i}^{j} p_s$

$$C[i,j] = \min_{i \le k \le j} \{ p_k \cdot 1 + C[i,k-1] + \sum_{s=1}^{k-1} p_s (\text{level of } a_s \text{ in } T[i,k-1]) + \sum_{s=i}^{k-1} p_s + \sum_{s=k+1}^{j} p_s (\text{level of } a_s \text{ in } T[k+1,j]) + \sum_{s=k+1}^{j} p_s \}$$

$$\hookrightarrow C[k+1,j]$$

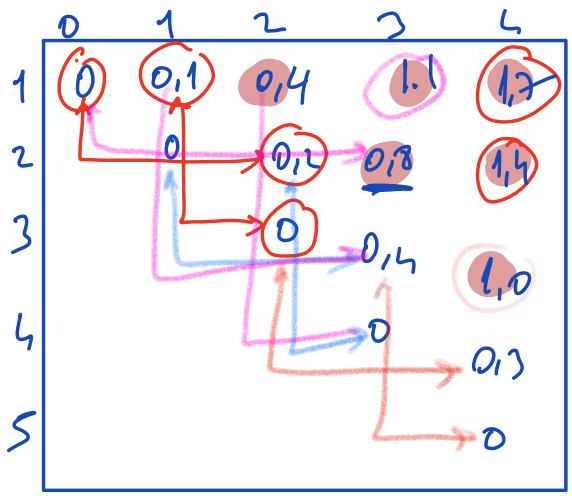
$$C[i,j] = \min_{i \le k \le j} \{ C[i,k-1] + C[k+1,j] \} + \sum_{s=i}^{j} p_s$$

$$C[i,i] = p_i$$

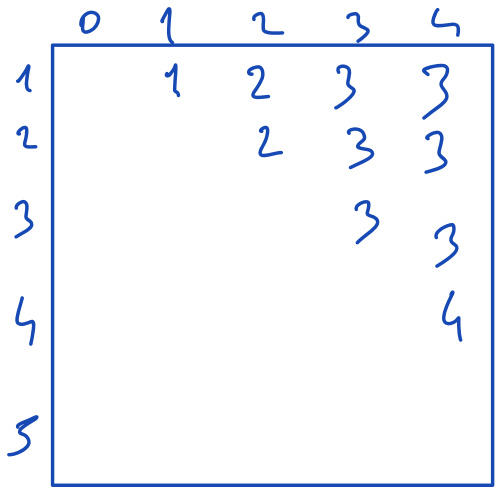


Example: key	A	B	C	D
probability	0.1	0.2	0.4	0.3

$$C[i,j] = \min_{i \le k \le j} \left[C[i,k-1] + C[k+1,j] \right] + \sum_{s=i}^{j} p_s$$



C[i,j]



Root matrix

$$C[1,2] = \min \begin{cases} C[1,0] + C[2,2] & k=1 \\ C[1,1] + C[3,2] & k=2 \end{cases} + (p_1+p_2) \quad 0,1+0,2$$

$$= \min\{ \underbrace{0+0,2}_{k=1}, \underbrace{0,1+0}_{k=2} \} + 0,3 = 0,4$$

$$C[2,3] = \min\{ \underbrace{0+0,4}_{k=2}, \underbrace{0,2+0}_{k=3} \} + (0,2+0,4) = 0,2+0,6 = 0,8$$

$$C[3,4] = \min\{ \underbrace{0+0,3}_{k=3}, \underbrace{0,4+0}_{k=4} \} + (0,4+0,3) = 0,3+0,7 = 1,0$$

$$C[1,3] = \min\{ \underbrace{0+0,8}_{k=1}, \underbrace{0,1+0,4}_{k=2}, \underbrace{0,4+0}_{k=3} \} + (0,1+0,4+0,3) = 1,1$$

$C[2,4] = \min\{\underbrace{0+1.0}_{L=2}, \underbrace{0,2+0.3}_{L=3}, \underbrace{0,8+0}_{L=4}\} + (0,2+0,4+0,3)$

$= 0,5 + 0,9 = 1,4$

$C[1,4] = \min\{\underbrace{0+1,4}_{L=1}, \underbrace{0,1+1.0}_{L=2}, \underbrace{0,4+0,3}_{L=3}, \underbrace{1,1+0}_{L=4}\} + 1$

$= 0,7 + 1 = 1,7$

$A =$

0	1	1	1	0	0	1
0	0	1	1	1	0	0
0	0	1	1	1	1	0
1	1	1	1	1	1	0
1	1	1	0	0	1	1
0	1	1	0	0	1	1

nxm array A.

Problem: To find the largest square of 1.

$A =$

1	1	1
1	1	0
1	1	1

$R =$

1	1	1
1	2	0
1	2	1

size of the largest square with bottom right index of (i,j)

$R[i,j] = 0$ if $A[i,j] = 0$

$R(i,j) = \min\{R(i-1,j), R(i,j-1), R(i-1,j-1)\} + 1$ if $A[i,j] = 1$

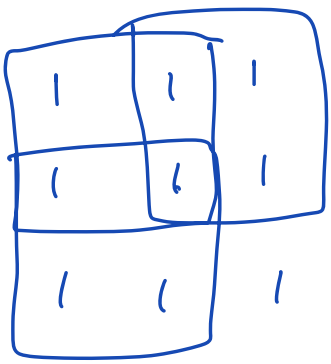
$R[1,j] = A[1,j]$ $R[i,1] = A[i,1]$

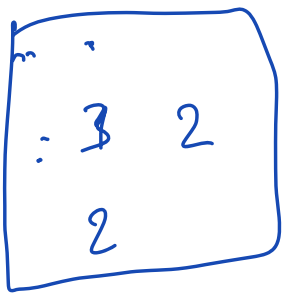
A

0	1	1	1	0	0	1
0	0	1	1	1	0	0
0	0	1	1	1	1	0
1	1	1	1	1	1	0
1	1	1	0	0	1	1
0	1	1	0	0	1	1

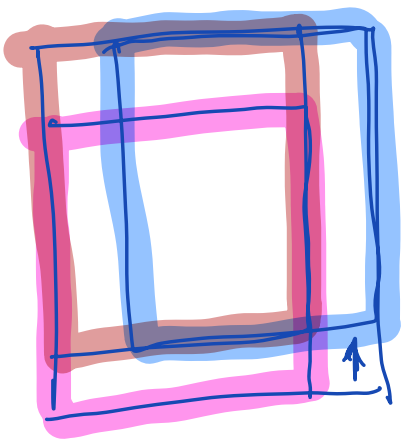
R

0	1	1	1	0	0	1
0	0	1	2	1	0	0
0	0	1	2	2	1	0
1	1	1	2	3	2	0
1	2	2	0	0	1	1
0	1	2	0	0	1	2



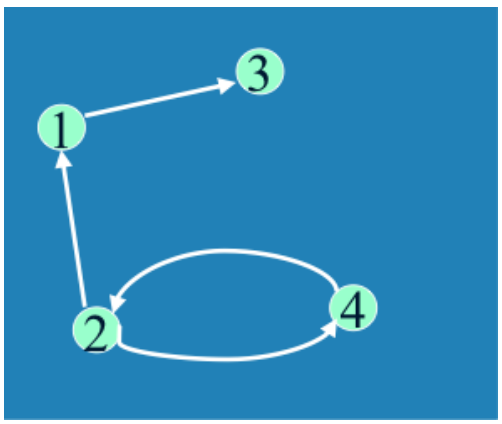


The max value you in the size & right bottom corner of the largest corner.



Warshall's alg.

$$R^{(k)}(i,j) = R^{(k-1)}(i,j) \vee \left(R^{(k-1)}(i,k) \wedge R^{(k-1)}(k,j) \right)$$



$R^{(0)} =$

0	0	1	0
1	0	1	1
0	0	0	0
0	1	0	0

$R^{(1)} =$

0	0	1	0
1	0	1	1
0	0	0	0
0	1	0	0

$R^{(2)} =$

0	0	1	0
1	0	1	1
0	0	0	0
1	1	1	1

$R^{(3)} =$

0	0	1	0
1	0	1	1
0	0	0	0
1	1	1	1

$R^{(4)} =$

0	0	1	0
1	1	1	1
0	0	0	0
1	1	1	1